2 Inorganic chemistry of group 7 (limited to chlorine, bromine and iodine)

a recall the characteristic physical properties of the elements limited to the appearance of solutions of the elements in water and hydrocarbon solvents

b describe and carry out the following chemical reactions of halogens:

- i oxidation reactions with metal and non-metallic elements and ions such as iron(II) and iron(III) ions in solution
- ii disproportionation reactions with cold and hot alkali, e.g. hot potassium hydroxide with iodine to produce potassium iodate(V)

c carry out an iodine/thiosulfate titration, including calculation of the results and evaluation of the procedures involved, e.g. determination of the purity of potassium iodate(V) by liberation of iodine and titration with standard sodium thiosulfate solution

d describe and carry out the following reactions:

- i potassium halides with concentrated sulfuric acid, halogens and silver nitrate solution
- ii silver halides with sunlight and their solubilities in aqueous ammonia solution
- iii hydrogen halides with ammonia and with water (to produce acids)

e make predictions about fluorine and astatine and their compounds based on the trends in the physical and chemical properties of halogens.

2.8 Kinetics

Students will be assessed on their ability to:

a recall the factors that influence the rate of chemical reaction, including concentration, temperature, pressure, surface area and catalysts

b explain the changes in rate based on a qualitative understanding of collision theory

c use, in a qualitative way, the Maxwell-Boltzmann model of the distribution of molecular energies to relate changes of concentration and temperature to the alteration in the rate of a reaction

d demonstrate an understanding of the concept of activation energy and its qualitative relationship to the effect of temperature changes on the rate of reaction

e demonstrate an understanding of the role of catalysts in providing alternative reaction routes of lower activation energy and draw the reaction profile of a catalysed reaction including the energy level of the intermediate formed with the catalyst

f carry out simple experiments to demonstrate the factors that influence the rate of chemical reactions, e.g. the decomposition of hydrogen peroxide.